GenAI Opportunities Report

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CERN

Geneva

Organization overview

CERN, the European Organization for Nuclear Research in Geneva, runs the Large Hadron Collider (LHC) and is funded mainly by its 23 Member States (2023 budget about CHF 1.23 billion; about 3,560 employed personnel plus roughly 13,400 associated members, per CERN's 2023 brochure). In November 2025 its Enlarged Directorate approved an organisation-wide AI strategy covering scientific discovery, productivity, talent and partnerships.

Your request: "Solve the theory of everything." **Candid framing:** no GenAI tool can currently deliver a verified theory of everything, and CERN's own Maurizio Pierini stresses that AI cannot replace the experiments themselves. The realistic opportunity is to speed up the work that surrounds that goal: theory exploration, equation discovery and searching data for physics beyond the Standard Model.

Three GenAI opportunities

01 Derivation and literature copilot for theorists

Use a frontier LLM with retrieval over arXiv/INSPIRE and internal notes to summarise competing unification approaches, cross-check derivations step by step, and flag where assumptions conflict. Humans stay responsible for every claim; outputs are treated as leads to verify.

> **EXAMPLE** (illustrative) *Input:* "List the assumptions shared by two named grand-unification proposals, cite the papers, and re-derive the key gauge-coupling relation, showing each step." *Output:* a cited comparison table and a derivation with each step marked checked/unchecked. Estimated saving: a few hours per literature review (estimate, not measured).

02 LLM-guided equation discovery on experimental data

LLM-SR (arXiv 2404.18400) treats equations as programs: an LLM proposes equation skeletons using scientific knowledge, then parameters are fitted to data and the best hypotheses guide the next round. The paper reports that it outperforms state-of-the-art symbolic regression on four problems in three scientific domains (not particle physics specifically). A first test could be a known, well-understood relationship.

> **EXAMPLE** (illustrative) *Scenario:* give the method a published invariant-mass or cross-section dataset with a known formula held back; check whether it recovers the formula and generalises to a held-out energy range. Success criterion defined before the run.

03 Anomaly-to-hypothesis assistant for new physics searches

CERN already uses anomaly detection on collision data (similar to fraud detection) to look for patterns that deviate from existing theory. A GenAI layer could turn flagged anomalies into a readable brief: candidate explanations from the literature, relevant models, and suggested follow-up checks.

> **EXAMPLE** (illustrative) *Input:* a statistics summary of an excess in one dijet mass window. *Output:* a one-page brief listing known detector or background explanations first, then 3 beyond-Standard-Model candidates with references, for physicists to triage. Estimated triage time cut from days to hours (estimate).

Practical next steps

1. **This week:** run opportunity 01 as a pilot with 3 to 5 theorists on one open question, logging every error the model makes.
2. Replicate LLM-SR on a public dataset with a known answer to measure value before touching open problems.
3. Align with the November 2025 CERN AI strategy owners so pilots follow its governance and partnership routes.
4. Set rules up front: independent verification of all model claims; data sharing per CERN policy.

Sources

CERN Brochure 2023: https://home.web.cern.ch/sites/default/files/2023-09/CERN-Brochure-2023-002-Eng.pdf

Quantum Zeitgeist: CERN approves AI strategy: https://quantumzeitgeist.com/cern-ai-ai-strategy/

Swissinfo: At CERN, AI will drive future discoveries: https://www.swissinfo.ch/eng/research-frontiers/at-cern-ai-will-drive-future-discoveries/91723774

LLM-SR (arXiv 2404.18400): https://arxiv.org/html/2404.18400v1

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